Effect of Microgravity on the Pharmacokinetics of Ibuprofen in Humans

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The pharmacokinetics of ibuprofen were studied under microgravity (μG) conditions and compared with those at normal gravity (1G) in humans. Six healthy human volunteers were given 600 mg oral dose ibuprofen during 1-day simulated μG antiorthostatic bed rest position, then at normal position (1G) in a sequential design with 7 days washout time. Saliva and plasma samples were obtained up to 8 hours after dosing. Ibuprofen was not detected in all saliva samples. Pharmacokinetic parameters in plasma were calculated by either noncompartmental analysis or a 1-compartment model using the Kinetica program. Absorption kinetic parameters were then predicted by ADAM and PE modules using the Simcyp program. Results have showed an increased rate of ibuprofen dissolution and absorption and hence faster onset of action under μG conditions. However, rate of drug elimination and bioavailability was not affected by μG, suggesting no need for dose adjustment.

Keywords: *Microgravity; ibuprofen; pharmacokinetics; antiorthostatic bed rest position; Simcyp*

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Drug pharmacokinetics differs significantly at microgravity (μG) conditions in space as compared with those at normal gravity (1G) on Earth. Pharmacokinetics involves 4 main biological processes: absorption, distribution, metabolism, and excretion (ADME). Several ground-based studies were done to simulate μG on Earth or to find noninvasive methods to study ADME in space. ADME models used for pharmacokinetics at 1G will be modified to work at μG. This can be achieved by incorporation of needed equations and factors to predict ADME properties in space from ADME properties on Earth. For example, drug absorption from the gastrointestinal tract depends on many factors such as gastric emptying rate, intestinal motility, blood perfusion, and mean residence time. Gastric emptying rate depends on volume of the meal, temperature of the food, body position, and also Earth gravity. Similarly, intestinal motility, blood perfusion, and mean residence time will also change at μG.[1-9]

ADME properties of drugs are affected mainly because of physiological changes, under μG conditions, such as fluid redistribution, fluid loss, electrolyte imbalance, muscle damage, and cardiovascular changes. In microgravity, fluid pressure equalizes, and fluids shift toward the head. The brain interprets the increase of fluid in the cephalic area as an increase in total fluid volume. In response, it activates excretory mechanisms, causing calcium loss and bone demineralization and formation of urinary stones. In addition, fluid redistribution leads to potassium and sodium imbalances and disturbs the autonomic regulatory system. Blood volume may decrease by 10%, causing cardiovascular reconditioning such as lower diastolic pressure, loss of red blood cells, and a tendency to spontaneously faint. Also, heart stroke volume and cardiac output are generally elevated during space flights. Moreover, distribution and efficacy of immune agents are affected, leading to an increase in neutrophils and a decrease in eosinophils, monocytes, and B cells, as well as a rise in steroid hormones and damage to T cells. The first signs of exposure to a zero-gravity atmosphere include symptoms such as headache, nausea, vomiting, congestion, back pain, sleeplessness, and space motion sickness.

Upon return to gravity, fluid is pulled back to the lower extremities, and the cardiac output falls to subnormal levels. It can take several weeks for such changes to return to normal functions. Muscle atrophy is caused by the lack of their use, leading to contractile protein loss, tissue shrinkage, and weight loss. Motion sickness also occurs to crew members because

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of spatial disorientation at µG conditions. Even bacterial cell membranes tend to be thicker and less permeable, reducing the efficacy of antibiotics. On the other hand, it was found that µG conditions have caused an insignificant decrease in total CYP450 liver enzymes. Attempts were done to reduce the above problems by using artificial gravity or acceleration such as the research done using the Rotating Space Station Simulator at the NASA Langley Research Center. Nevertheless, physiological changes still exist at µG conditions and need to be dealt with as a fact.[1-9]

OBJECTIVES

The objective is to compare plasma and saliva pharmacokinetics under 1G with those under simulated µG conditions. The anti-inflammatory drug ibuprofen was used as a model drug since no previous ground studies were done on it. The results can help us understand more the effects of µG conditions on ADME and hence be able to correct dose regimens for drugs to be used in flight by crew members to treat muscle and back/neck pain.

EXPERIMENTAL

Study Design

The pharmacokinetics of 600 mg oral dose ibuprofen in 6 healthy human volunteers during 1-day simulated µG antiorthostatic bed rest position (ABR) were compared with the pharmacokinetics at normal position (1G) in a sequential design with a 7-day washout time. The study was approved by the institutional review board and Jordan Food and Drug Administration and conducted by the Jordan Center for Pharmaceutical Research at Al-Mowasah Hospital as per the International Conference on Harmonization, good clinical practice, and Declaration of Helsinki guidelines.

After a 10-hour overnight fast, a 600-mg oral dose of ibuprofen tablets was given with 240 mL water. Blood and saliva samples were withdrawn at 0, 0.25, 0.5, 0.75, 1, 1.33, 1.66, 2, 2.5, 3, 4, 5, 6, and 8 hours postdose. Samples were deep stored at −20°C until assayed by a validated high-performance liquid chromatography method, with a linear range of 0.5 to 30 µg/mL (Figure 1).

Sample Size Determination

Sample size is based on a previous ibuprofen intersubject pharmacokinetic variability of 25%. The minimum sample size needed with 80% power and 95% confidence was 6 participants.

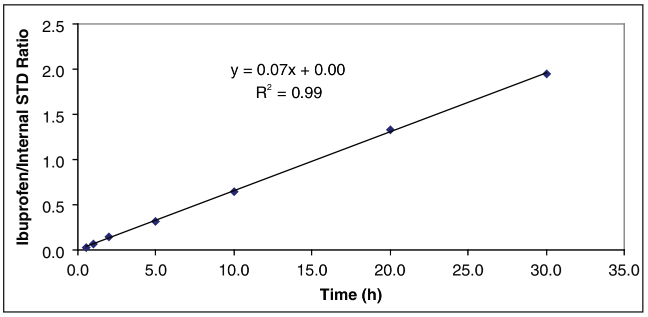

Figure 1. Ibuprofen standard curve.

Human Participants

Six men, aged 18 to 45 years and with a body mass index between 18.5 and 24.9 kg/m^2, were enrolled. Medical history, vital signs, physical examination, electrocardiogram, and laboratory safety test results had no evidence of a clinically significant deviation from a normal medical condition as evaluated by the clinical investigator. Participants passed the ABR tolerability test (able to eat and urinate while in ABR position) and signed the informed consent form before initiation of the study.

DATA ANALYSIS

Pharmacokinetic Analysis

Pharmacokinetic parameters for drug concentration were calculated by noncompartmental analysis (NCA) using the Kinetica 2000 computer program. Pharmacokinetic parameters were area under the concentration curves to 8 hours and to infinity ($AUC_{0\rightarrow t}$, $AUC_{0\rightarrow\infty}$), maximum measured plasma concentration and its time (C_{max} and t_{max}), first-order elimination rate constant (K_{el}), and half-life ($t_{1/2}$). However, the absorption rate constant (K_a) was calculated by data fitting using a 1-compartment model.

Statistical Analysis

Analyses of variance (ANOVAs) were performed, assuming normal distribution, on log-transformed pharmacokinetic parameters of $AUC_{0\rightarrow\infty}$, $AUC_{0\rightarrow t}$, C_{max}, K_{el}, K_a, and $t_{1/2}$. However, t_{max} was compared using the nonparametric Friedman test with chi-square distribution. A 5% level of significance was used for all statistical comparisons.

Table I Ibuprofen Plasma Levels (μG/mL) After a 600-mg Oral Dose to 6 Healthy Volunteers Under Normal Gravity (1G) and Microgravity (μG) Body Positions

Position	1G						μG					
Time, h	Vol. 1	Vol. 2	Vol. 3	Vol. 4	Vol. 5	Vol. 6	Vol. 1	Vol. 2	Vol. 3	Vol. 4	Vol. 5	Vol. 6
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.25	0.43	0.00	0.91	1.57	2.14	11.62	16.10	30.77	0.00	1.74	10.67	8.82
0.50	4.22	0.84	10.92	10.44	12.62	26.13	45.83	38.15	25.26	23.34	24.22	19.98
0.75	8.16	3.39	21.15	41.73	38.95	28.09	48.32	34.15	35.63	39.44	34.18	18.29
1.00	13.73	23.64	33.90	48.65	49.10	26.96	43.52	32.66	32.93	43.15	42.43	17.78
1.33	36.06	29.29	36.55	45.01	45.14	26.13	31.48	28.48	29.91	42.32	42.70	15.04
1.66	39.24	32.18	29.61	42.80	38.31	27.85	25.14	26.68	24.84	38.06	32.34	13.55
2.00	35.07	36.45	24.71	34.75	33.41	27.84	21.40	25.22	18.73	31.39	29.67	11.38
2.50	27.77	39.10	18.25	26.80	29.25	24.61	15.95	22.48	14.71	25.80	21.46	24.77
3.00	26.12	29.73	14.28	20.70	21.86	18.60	13.11	18.48	12.94	20.68	16.45	23.97
4.00	14.97	19.38	8.87	15.01	15.10	11.49	8.55	13.32	7.28	14.85	11.46	18.58
5.00	9.50	13.41	5.22	10.46	9.30	6.81	5.67	9.36	4.54	9.54	7.51	10.90
6.00	5.20	8.44	3.20	6.73	6.61	4.36	3.62	6.13	2.61	6.90	4.78	6.94
8.00	2.58	4.52	1.40	3.30	3.72	1.89	1.71	3.58	1.06	3.57	2.91	3.68

Table II Ibuprofen Mean (CV%) Plasma Pharmacokinetic Parameters After a 600-mg Oral Dose to 6 Healthy Volunteers Under Normal Gravity (1G) and Microgravity (μG) Body Positions

Body Position/Pharmacokinetic Parameter	1G, Mean (CV%)	μG, Mean (CV%)	*P* Value (Power %)
$AUC_{0\rightarrow t}$, μg·mL/h	20.97 (17)	112.66 (16)	.49 (90)
$AUC_{0\rightarrow\infty}$, μg·mL/h	128.88 (18)	120.77 (17)	.57 (92)
C_{max}, μg/mL	40.12 (20)	38.79 (21)	.77 (94)
$t_{1/2}$, h	1.82 (15)	1.96 (16)	.47 (89)
K_{el}, h^{-1}	0.39 (13)	0.36 (18)	.47 (89)
t_{max}, h	1.37 (46)	1.14 (64)	<.05
K_a, $h^{-1,a}$	0.79 (35)	2.24 (96)	.03 (63)

[a]Apparent absorption rate constant was determined by 1-compartment fitting.

Absorption Kinetics and Dimensional Analysis

Absorption kinetic parameters—effective intestinal permeability (Peff) and segmental fraction dose absorbed (Fa)—were calculated by deconvolution of plasma data using the ADAM model in the Simcyp program, version 9.3. Parameter estimation for data fitting was performed using the combined Nelder-Mead minimization algorithms.

RESULTS AND DISCUSSION

Plasma ibuprofen levels and pharmacokinetic parameters are presented in Tables I and II. Figure 2 shows average plasma levels of ibuprofen under both positions. However, ibuprofen was not detected in the saliva in all participants, and hence salivary pharmacokinetic parameters were not obtained.

All pharmacokinetic parameters except t_{max} showed insignificant differences at the .05 level of significance between the normal gravity position and the ABR position. This showed that bioavailability and the elimination process were not changed under the μG position. In addition, they showed low variability and high statistical power, indicating adequate study design.

On the contrary, t_{max} and K_a were significantly different at the ABR position and showed high intersubject variability, reaching 96%. The physiological changes of fluid shift to the upper trunk of the body and also slower gastric emptying and intestinal motility, while under the μG position, have caused

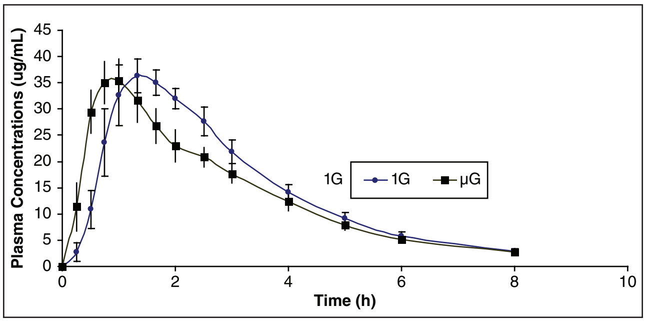

Figure 2. Ibuprofen mean (SE) plasma levels (μg/mL) after a 600-mg oral dose to 6 healthy volunteers under normal gravity (1G) and microgravity (μG) body positions.

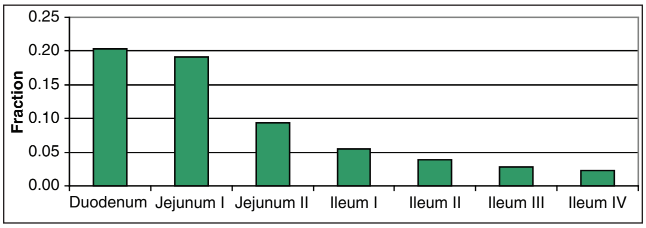

Figure 3. Regional distribution of the fraction of ibuprofen oral dose absorbed.

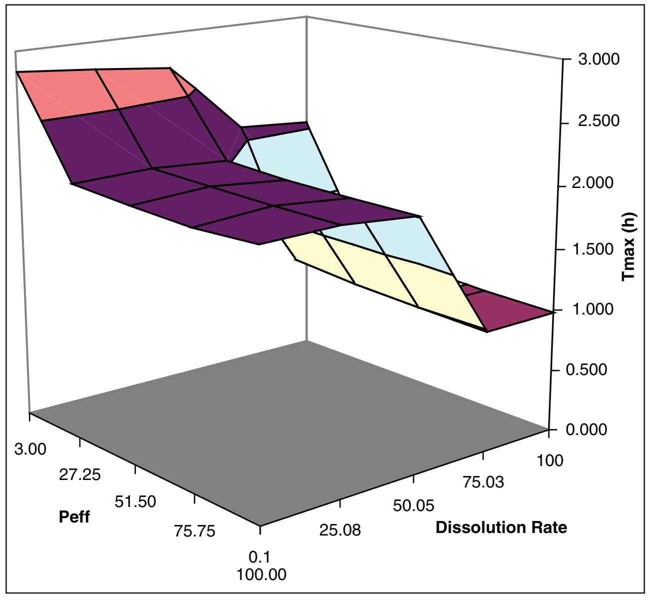

Figure 4. Sensitivity analysis of t_{max} as a function of effective intestinal permeability and dissolved rate.

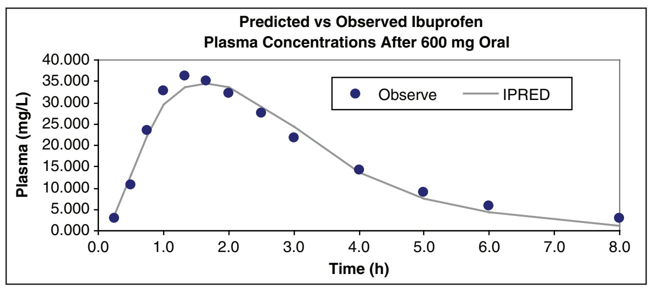

Figure 5. In silico prediction and observed profiles of ibuprofen.

faster ibuprofen dissolution and absorption. The increased residence time in the gut environment under the μG position gave adequate time for drug dissolution. This led to larger apparent K_a and smaller t_{max} values, as shown in Table II. The clinical implication of such differences in drug pharmacokinetics is faster onset of ibuprofen action under μG conditions, yet with similar elimination and bioavailability, suggesting no need for dose adjustment. This was in agreement with what was concluded previously[10,11] that ibuprofen has low water solubility, and its absorption is controlled by dissolution in vivo. In addition, ibuprofen has high intestinal permeability and is considered a class II drug according to the new biopharmaceutics classification system.[12] Hence, the absorption rate constant was considered as apparent since it reflects a dissolution process rather than an absorption process.

Optimized intestinal permeability values (Peff × 10^{-3} cm/s) were 4.41 and 4.83 for the 1G and μG positions, respectively. This has led to similar segmental fractions of dose absorbed (Fa) as in Figure 3, with duodenum and jejunum I having the highest fractions. This confirms the above suggestion that μG position has led to an increase in dissolution and apparent absorption rates. However, as demonstrated by in silico simulations, intestinal permeability and fraction absorbed values are similar. The sensitivity analysis shows in Figure 4 that t_{max} was not changing at and above the calculated permeability values but was highly dependent on the dissolution rate above 50%. Representative plots are shown in Figures 5 and 6 showing accurate prediction (R^2 = 0.98).

CONCLUSIONS

The results of this research have showed an increased rate of ibuprofen dissolution and absorption and hence faster onset of action under μG conditions. However, the rate of drug elimination and bioavailability was not affected by μG, suggesting no need for dose adjustment. This outcome cannot be generalized to all drugs. More work is needed for class I

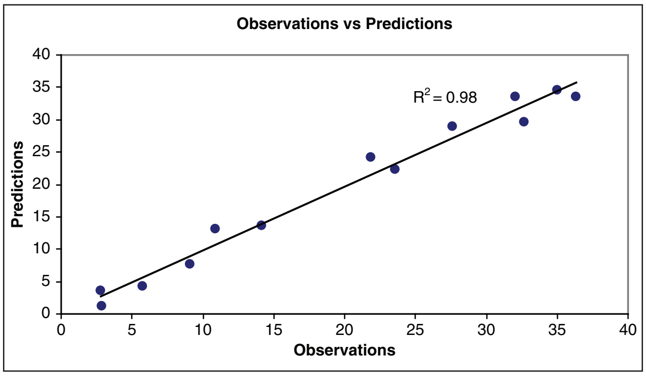

Figure 6. In silico prediction and observed data correlation.

drugs (highly soluble and highly permeable), where absorption and onset of action depend on gastric emptying.

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